

Part A · Multiple Choice

2 points each

1. **a) About 3%**

p. 243 — the oceans hold 97% and it is salty. Of the 3% that is fresh, two-thirds is locked in ice and one-third is in the ground.

2. **c) dendritic**

p. 247 — trellis patterns form on folded or tilted sedimentary rock, radial patterns on volcanoes and domes, and rectangular patterns where fractures or faults form a right-angled network.

3. **d) fastest on the outside of the bend**

p. 248 — the water on the outside has farther to travel. Friction also makes flow slowest along the bed and edges and fastest near the surface in the middle.

4. **b) a crescent-shaped depositional feature on the inside of a stream bend**

p. 248 — the matching erosional feature on the outside of the bend is the cut bank, shaped like a small cliff. Together they make a meander migrate across its floodplain.

5. **a) a stream overtops its banks, the velocity drops sharply, and sediment is deposited near the channel edge**

p. 253 — a stream reaches its highest velocity at the bank-full stage. Once it spills onto the wide floodplain the water has far more area to flow through, so velocity and carrying capacity collapse.

6. **b) small sand grains between 0.2 mm and 0.5 mm**

p. 252 — anything smaller or larger needs a higher velocity. Clay and silt grains stick strongly to one another, making them surprisingly hard to lift off the stream bed.

Part B · True or False

1 point each

7. **TRUE**

p. 252 — a 0.01 mm silt particle needs about 60 cm/s to be eroded, while a 1 mm sand grain needs only about 20 cm/s. Filip Hjulström worked this relationship out in the 1940s.

8. **FALSE**

That describes youthful streams. Old-age streams have very low gradients, rich meanders, oxbow lakes, broad floodplains, and low erosive energy. (pp. 249, 253)

9. **Youthful streams cut deep, straight V-shaped valleys with steep, ungraded gradients and many rapids and waterfalls. Mature streams widen their valleys, deposit sediment, and become more graded. Old-age streams occupy wide sediment-filled valleys among rolling hills and meander freely.**

Rubric: 1 pt each stage. **Youthful** — follows tectonic uplift; rapid erosion, deep V-shaped valleys, relatively straight paths, high gradients, ungraded profiles, rapids and waterfalls, often step-pool morphology. **Mature** — wider valleys, thick sediment deposits, gradients slowly reduced, grading increases, a small floodplain and early meanders or braiding appear. **Old age** — rolling hills, wide sediment-filled valleys, very low gradient, pronounced meanders, oxbow lakes and meander scars, deltas and levees. Credit noting that Davis worked before plate tectonics and without knowledge of glacial erosion, so parts of the model are dated but still useful. (p. 248)

10. **Traction, saltation, suspension, and dissolved load.**

Rubric: 0.75 pt each. **Traction** — large bedload particles pushed along the bottom by the force of the flow. **Saltation** — particles bouncing along the bed; these first two move the largest clasts, often only during floods. **Suspension** — smaller particles held up within the flowing water, especially where flow is turbulent; turbulent flow keeps sediment suspended far better than laminar flow. **Dissolved load** — ions such as calcium and chloride carried in solution, averaging about 15% of the total mass transported. (pp. 251–252)